

POPURI SUSHANTH

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📍 Villa 97, GreenCity, Visakhapatnam

PROJECTS

Portlio

Developed a full-stack SaaS platform that automatically generates and updates developer portfolios from GitHub repositories. Implemented GitHub OAuth authentication, Automated Screenshots, responsive portfolio templates, and subscription management using Next.js, TypeScript, Supabase, and Razorpay.

GeoPopulation Explorer

Developed an interactive demographic visualization platform using a 3D globe and real-time population pyramids to explore global population trends from 1950–2100. Built a Node.js data pipeline to process and optimize United Nations datasets into a high-performance JSON format, enabling fast client-side simulations.

Quantum-Classical Financial Fraud Detection

Developed a hybrid quantum-classical fraud detection system integrating a PyTorch neural network with a 6-qubit variational quantum circuit using PennyLane. Implemented data preprocessing with feature scaling, PCA, and class-weighted training for highly imbalanced financial datasets.

AI-Legislative Analyser

Developed a full-stack RAG-based chatbot for querying government policies and legal documents using FastAPI. Implemented automated document routing and context retrieval, integrated ScaleDown.ai to compress retrieved context by up to 92% for lower API costs and faster responses, and built a real-time analytics dashboard to monitor token usage and savings.

OTHER PROJECTS

GENZ-Projects

Cognitive Stress Load Monitoring

Typing Test

Password Strength Visualizer

Multi Functional Calculator

EDUCATION

GITAM University - B.Tech Computer Science	(2023–Present)	CGPA: 9.2
FIITJEE - Intermediate	(2021–23)	83%
SGS - Class X	(2020-21)	75%

INTERNSHIPS

Research Intern - NIT Calicut

Worked on mitigating crosstalk effects in quantum computers by investigating quantum circuit partitioning and optimization techniques. Implemented and evaluated partitioning algorithms including KaHyPar, METIS, and Branch-and-Bound using Qiskit and QDisLib, simulated realistic quantum noise models, and analyzed their impact on circuit fidelity, execution overhead, and scalability to improve reliable quantum computation on NISQ devices.

PUBLICATIONS

“Impact Of Quantum Computing on RSA Cryptographic Security”

Conducted a comparative study on RSA cryptographic security in the context of quantum computing by analyzing classical RSA performance, the computational complexity of Shor's algorithm and the General Number Field Sieve (GNFS), and the impact of quantum noise through simulation. The work highlights the gap between RSA's theoretical quantum vulnerability and the practical limitations of current quantum hardware, emphasizing the need for post-quantum cryptographic migration.

SKILLS

Languages

Python JavaScript Java C SQL

Frontend

React.js Next.js HTML5 CSS3

Backend

Node.js FastAPI REST APIs

Databases

Supabase MongoDB

Tools & Platforms

Git GitHub Vercel Netlify RazorPay

CERTIFICATIONS

- Meta Front-End Developer Professional Certificate
 - Meta Back-End Developer Professional Certificate
 - Google AI Essentials
 - Google Prompting Essentials
 - DeepLearning.AI – AI For Everyone
 - IBM Introduction to Artificial Intelligence
 - Anthropic - Model Context Protocol (MCP) Advanced
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PARTICIPATIONS

- Google TechSprint 2025 – Semi-Finalist
- SIH 2025 Participant
- TMCG Idea Sprint 3.0 Participant